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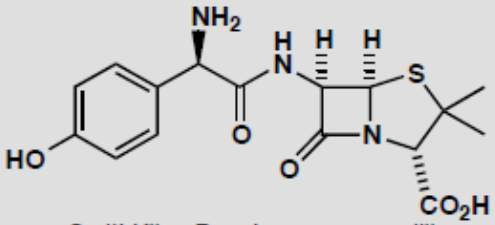
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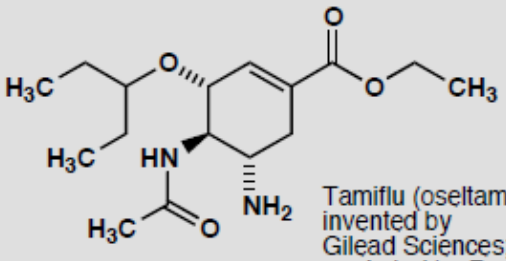
Tutorial 1

Problem 1

Draw for yourself the structures of amoxicillin and Tamiflu shown on page 10 of the textbook. Identify on your diagrams the functional groups present in each molecule and the ring sizes. Study the carbon framework: is there a single carbon chain or more than one? Are they linear, branched, or cyclic?

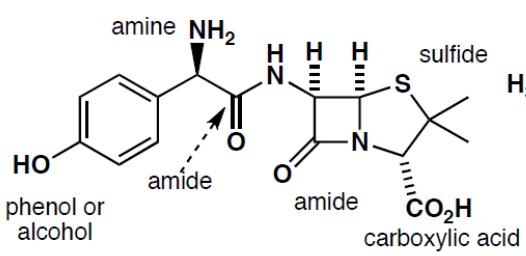


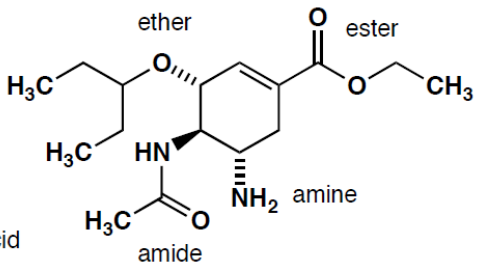
SmithKline Beecham's amoxycillin
β-lactam antibiotic
for treatment of bacterial infections

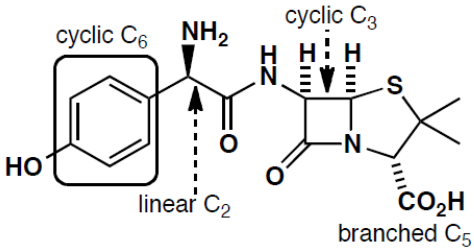


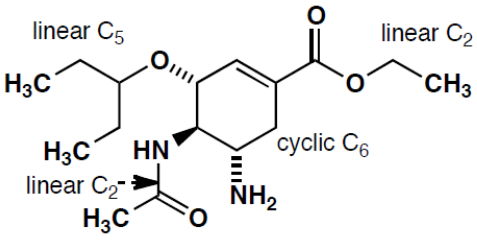
Tamiflu (oseltamivir):
invented by
Gilead Sciences;
marketed by Roche

The ring sizes are easy and we hope you noticed that one bond between the four- and the five-membered ring in the penicillin is shared by both rings.









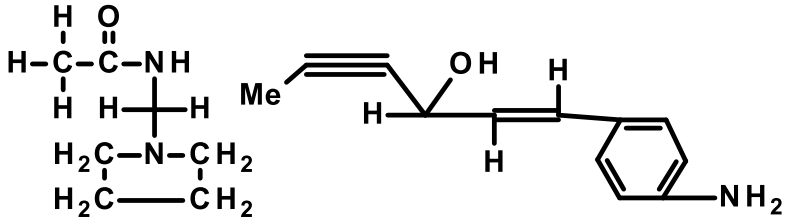
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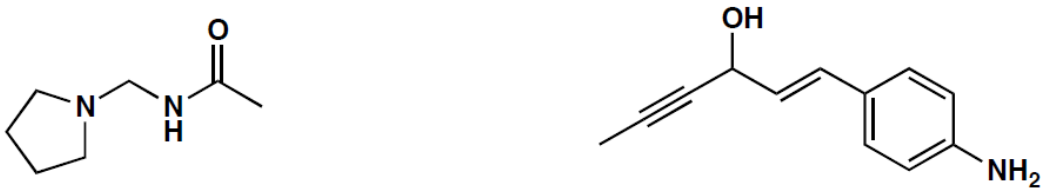
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Problem 2

What is wrong with these structures? Suggest better ways to represent these molecules.



The bond angles are grotesque with square planar saturated carbon atoms, bent alkynes with 120° bonds, linear alkenes with bonds at 90° or 180°, bonds coming off a benzene ring at the wrong angles and so on. If properly drawn, the left hand structure will be clearer without the hydrogen atoms. Here are better structures for each compound but you can think of many other possibilities.

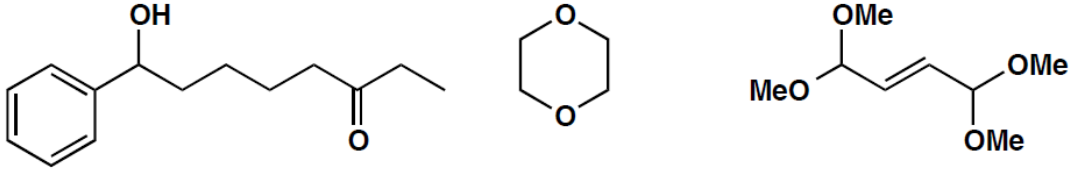


Problem 3

Translate these very poor structural descriptions into something more realistic. Try to get the angles about right and, whatever you do, don't include any square planar carbon atoms or any other bond angles of 90°.

(a) $C_6H_6CH(OH)CH_2)_4COC_2H_5$
(b) $O(CH_2CH2)_2O$
(c) $(CH_3O)\ CH{=}CHCH(OCH_3)_2$

$C_6H_5CH(OH)(CH_2)_4COC_2H_5$ $O(CH_2CH_2)_2O$ $(CH_3O)_2CH{=}CHCH(OMe)_2$

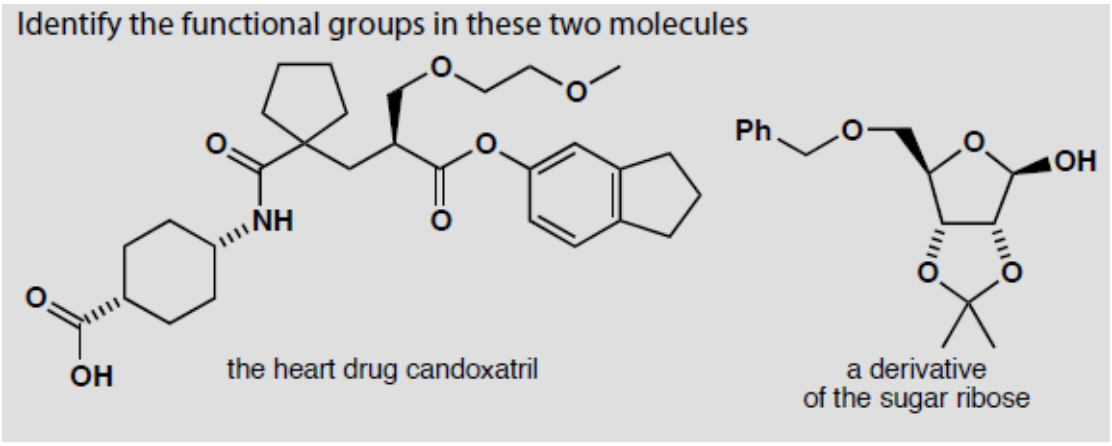


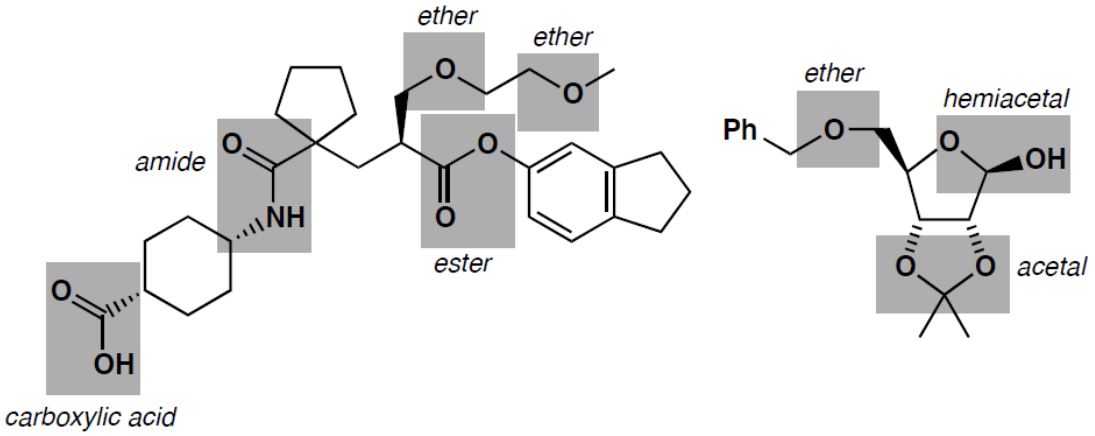
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Problem 4





Tutorial 2 (Answers)

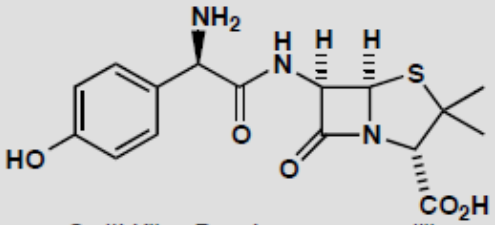
Problem 1

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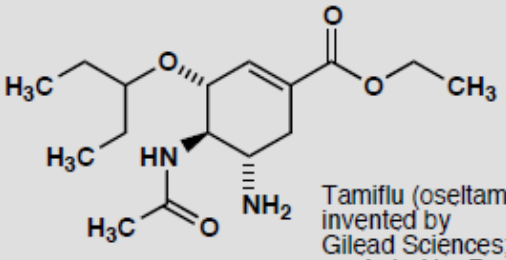
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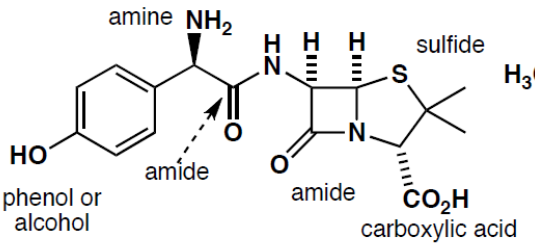


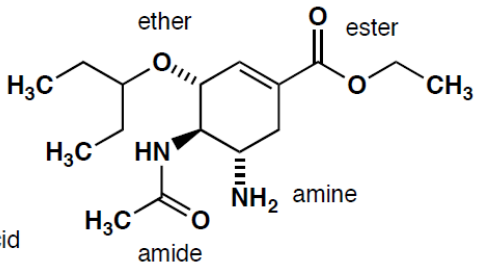
SmithKline Beecham's amoxycillin
β-lactam antibiotic
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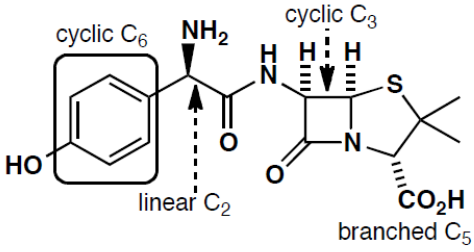


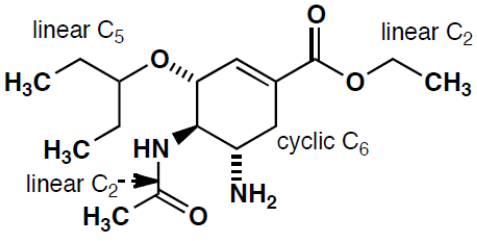
Tamiflu (oseltamivir):
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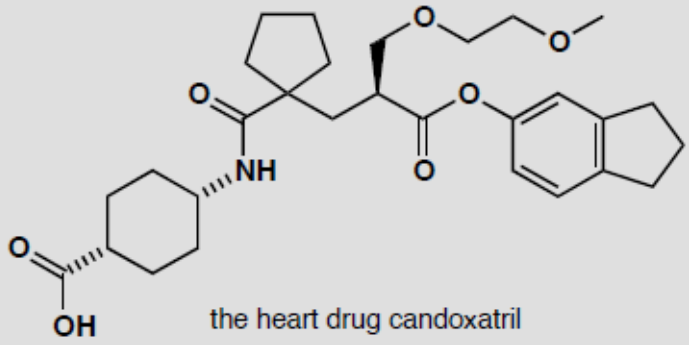
Problem 2

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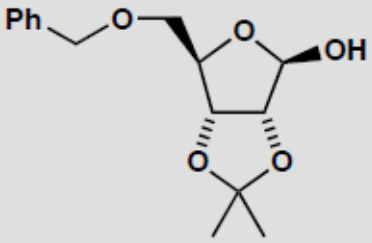
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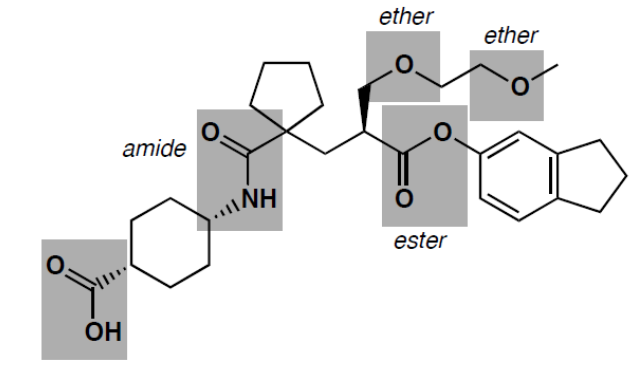
Identify the functional groups in these two molecules

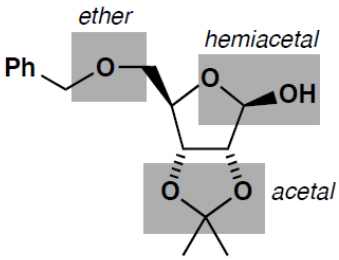


the heart drug candoxatril



a derivative
of the sugar ribose





Problem 3

Draw structures for the compounds named systematically here. In each case suggest alternative names that might convey the structure more clearly if you were speaking to someone rather than writing.

(a) 1,4-di-(1,1-dimethylethyl)benzene

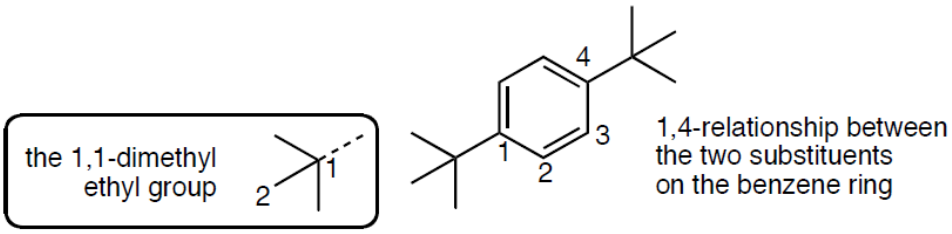
(b) 1-(prop-2-enyloxy)prop-2-ene

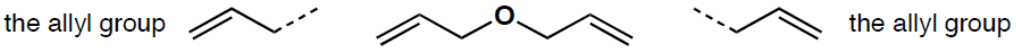
(c) cyclohexa-1,3,5-triene

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C) is benzene

Problem 4

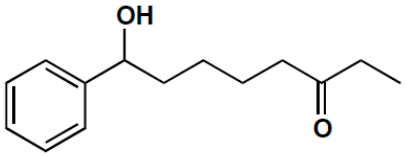
Draw full structures for these compounds, displaying the hydrocarbon framework clearly and showing all the bonds in the functional groups. Name the functional groups.

(a) $AcO(CH_2)_3NO_2$

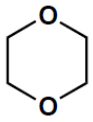
(b) MeO_2CCH_2OCOEt

(c) $CH_2{=}CHCONH(CH_2)_2CN$

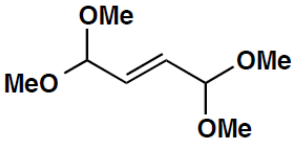
$C_6H_5CH(OH)(CH_2)_4COC_2H_5$



$O(CH_2CH_2)_2O$



$(CH_3O)_2CH{=}CHCH(OMe)_2$



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Problem 5

Draw structures for the folllowing molecules, and then show them again using at least one 'organic element' symbol in each.

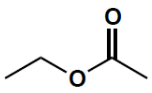
(a) ethyl acetate

(b) chloromethyl methyl ether

(c) pentanenitrile

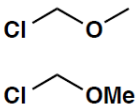
(d) *N*-acetyl *p*-aminophenol

(e) 2,4,6,-tri-(1,1-dimethylethyl)phenylamine

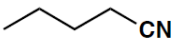


EtOAc

ethyl acetate

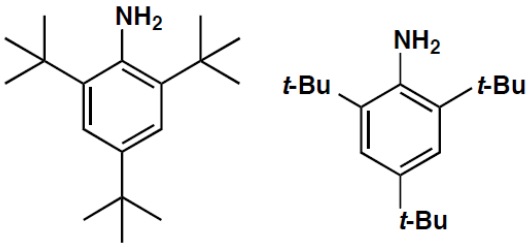


chloromethyl methyl ether

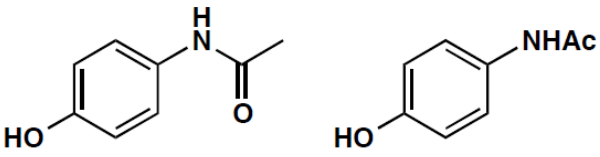


BuCN

pentanenitrile



2,4,6,-tri-(1,1-dimethylethyl)phenylamine



N-acetyl *p*-aminophenol

Tutorial 3

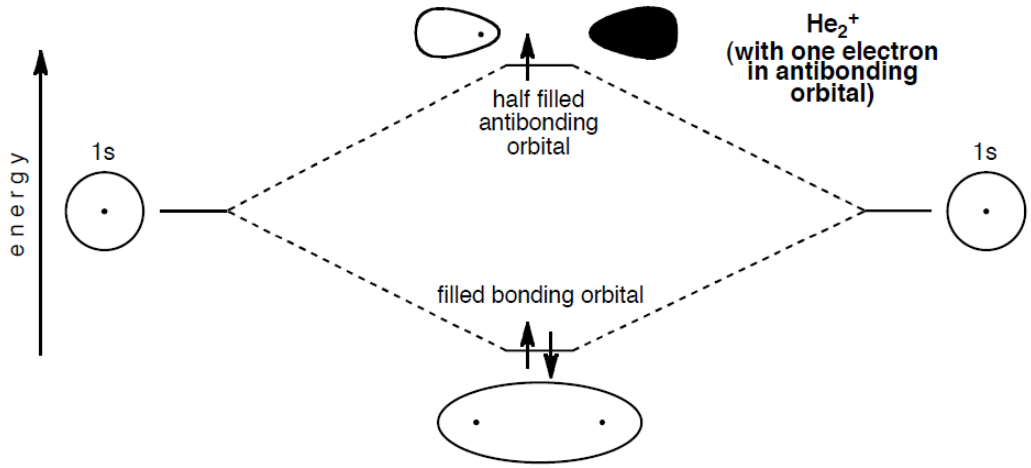
Problem 1

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Though the helium molecule He_2 does not exist (p.91 of your textbook explains why), the cation He_2^+ does exist. Why?

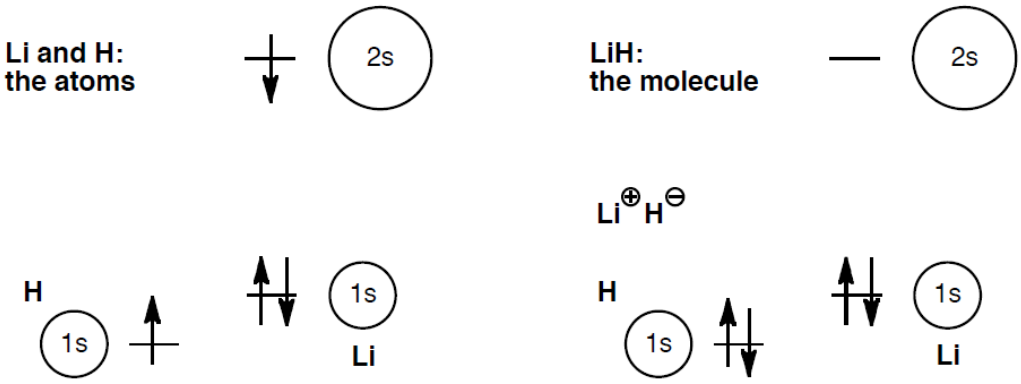


He_2 does not exist because the number of anti-bonding electrons is the same as the number of bonding electrons. The bond order is zero. But if we remove an electron from the diagram on p. 91 of the textbook we have He^{2+}, with two bonding electrons and only one anti-bonding electron. The bond order is one half. He^{2+} does exist.

Problem 2

Construct an Molecular Orbitals (MO) for LiH and suggest what type of bond it might have.

H has of course only one electron in a 1s orbital. Li has three – a full 1s shell and one electron in the 2s orbital. Li is very electropositive so its 2s orbital is high in energy—much higher than that of the 1s orbital of H. An electron is more stable in the 1s orbital of H than in the 2s orbital of Li, and the molecule is ionic. Both ions have the same electronic configuration: 1s2



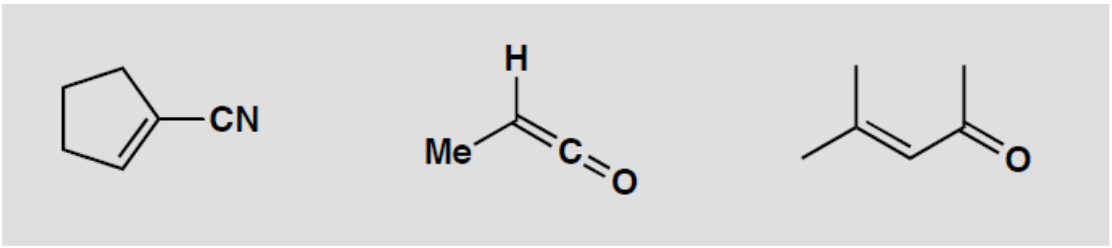
Problem 3

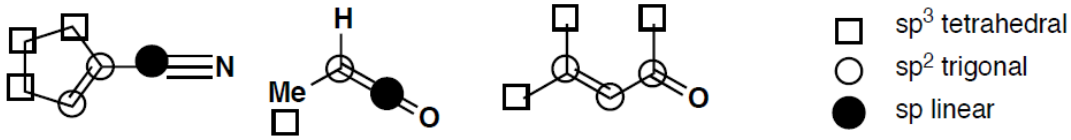
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What is the hybridization and shape of each carbon atom in these molecules?





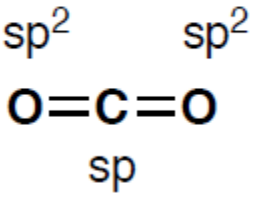
Problem 4

Draw detailed structures for these molecules and predict their shapes. We have deliberately made non-committal drawings to avoid giving away the answer to the question. Don't use these sorts of drawings in your answer.

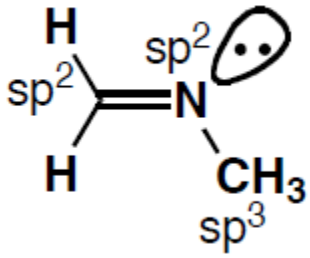
CO_2, $CH_2{=}NCH_3$, $CH_2{=}C{=}CH_2$, $(CH_2)_2O$

Carbon dioxide is linear as it has only two C–C σ-bonds and no lone pairs on C. The C atom must be sp hybridized and the only trick is to get the two π-bonds orthogonal to each other. They must be like that because the p orbitals on C used to make the two π-bonds are

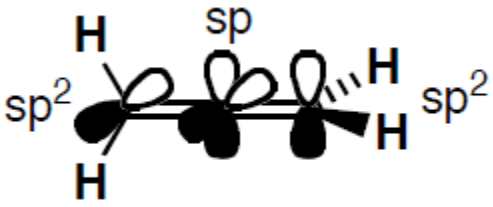
themselves orthogonal (px and py). Most people draw the O atoms as sp2 hybridized rather than sp or even unhybridized but this doesn't matter as you really can't tell.



The imine has a C=N double bond so it must have sp2 hybridized C and N. This means that the lone pair on nitrogen is in an sp2 orbital and not in a p orbital. The molecule is planar (except for the methyl group which is, of course, tetrahedral) and is bent at the nitrogen atom.



The next molecule $CH_2=C=CH_2$ is allene and it has the same shape as CO_2, and for the same reasons. We can now be sure that the end carbons are sp2 hybridized as they are planar, with the hydrogen substituents at 120°· to each other and to the rest of the molecule. As with CO_2, the two π bonds are orthogonal, meaning that the planes of the two terminal carbon atoms are also orthogonal, meaning that the molecule as a whole is not planar.

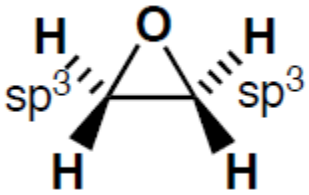


Finally, (CH2)2O must be a three-membered ring and therefore the C–C– O skeleton must be planar (three points are always in a plane!). The two carbon atoms are sp3 hybridized (four σ bonds) and are tetrahedral (though very distorted as the ring angle is 60°) with the H atoms above and below the ring. The oxygen atom is presumably also sp3 hybridized, but it's hard to tell experimentally.

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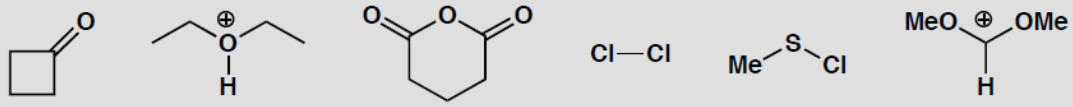
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Tutorial 4

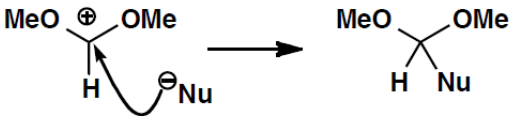
Problem 1

Each of these molecules is electrophilic. Identify the electrophilic atom and draw a mechanism for a reaction with a generalized nucleophile Nu^-, giving the structure of the product in each case.

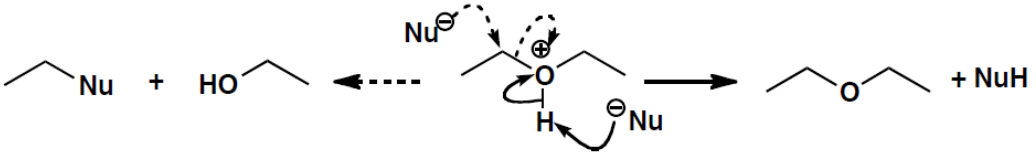


We have two cations, two carbonyl compounds and two compounds with σ bonds only.

One of the cations has three bonds to a positively charged carbon so that is the electrophilic site as it has an empty orbital. The nucleophile will attack here.



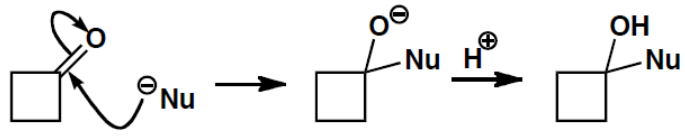
The other cation has a three-valent oxygen atom that cannot be the electrophilic site. The nucleophile must attack the proton instead. Some nucleophiles might attack the carbon atom joined to the cationic oxygen **(Very unlikely).**

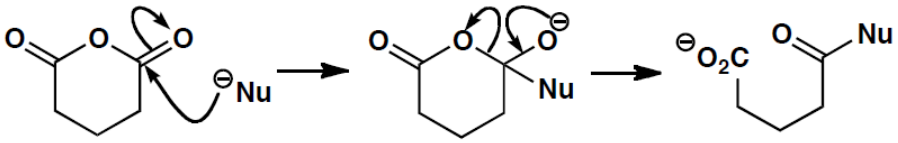


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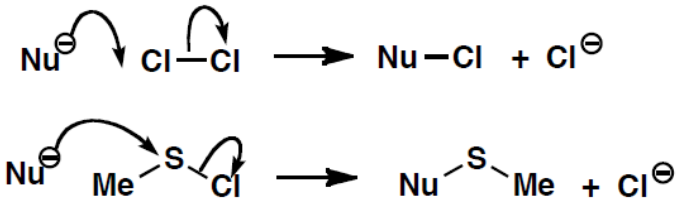
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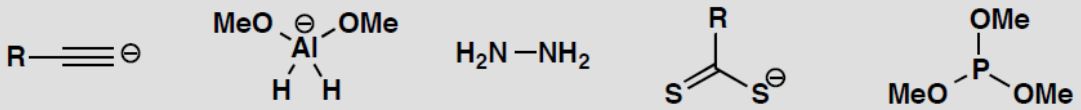


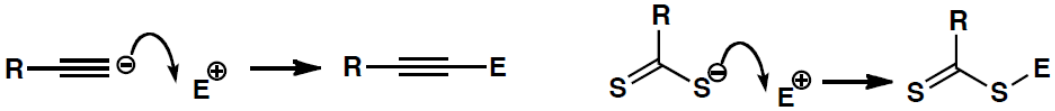
The two carbonyl compounds will be attacked at the carbonyl group by the nucleophile. In general, π-bonds are more easily broken than σ-bonds and the negative charge goes on to the electronegative oxygen atom.



Problem 2

Each of these molecules is nucleophilic. Identify the nucleophilic atom and draw a mechanism for a reaction with a generalized nucleophile E^+, giving the structure of the product in each case.

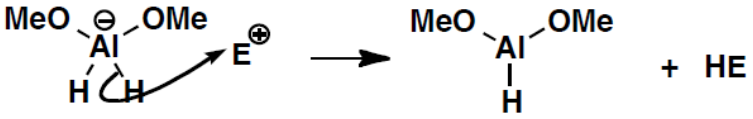


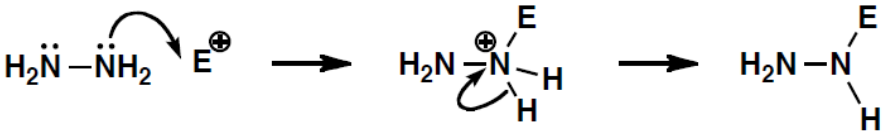


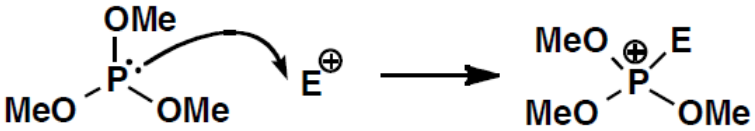
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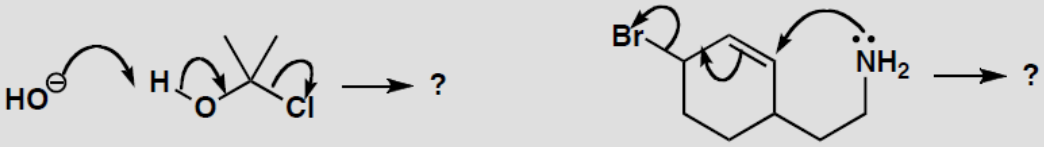


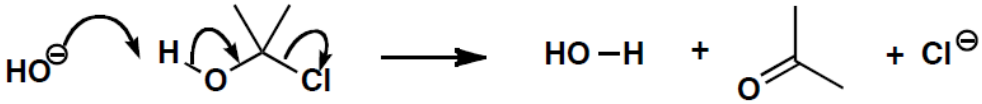


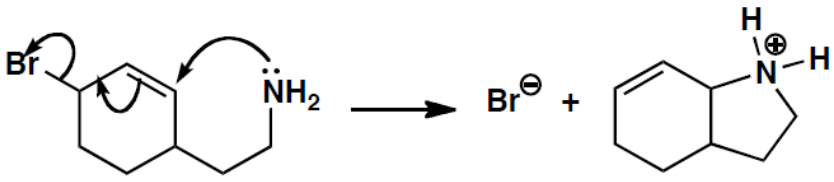


Problem 3

Complete these mechanisms by drawing the structure of the product(s).







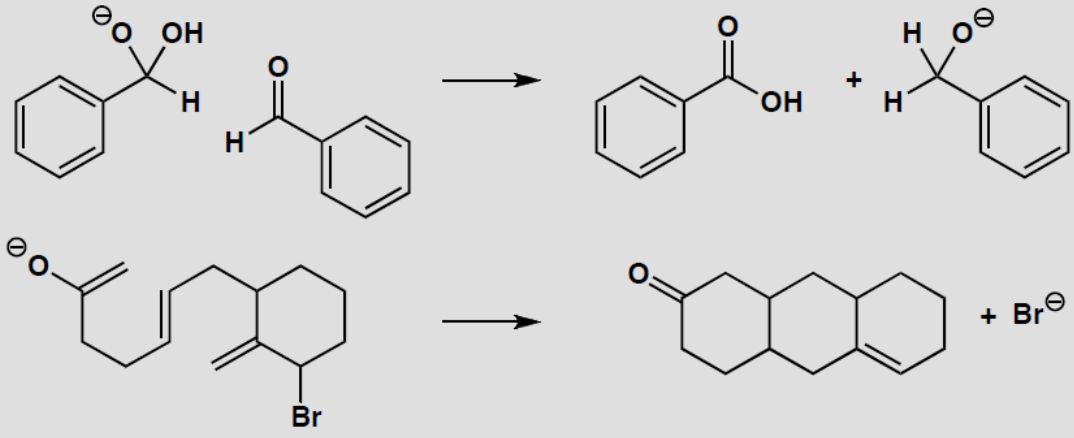
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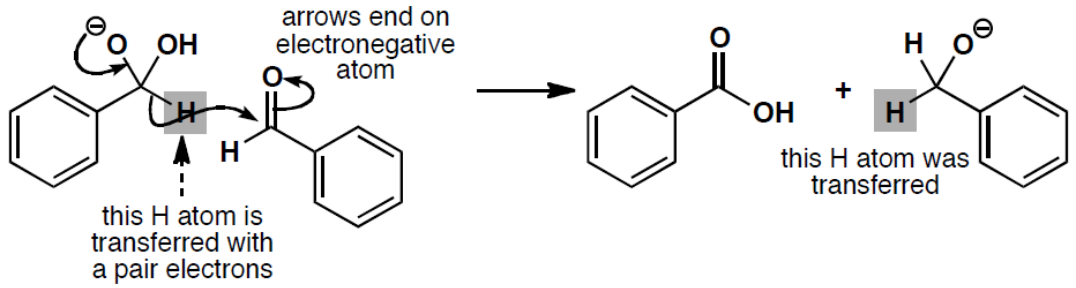
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Problem 4

Put in the curly arrows on these starting materials to show how the product is formed. The compounds are drawn in a convenient arrangement to help you.

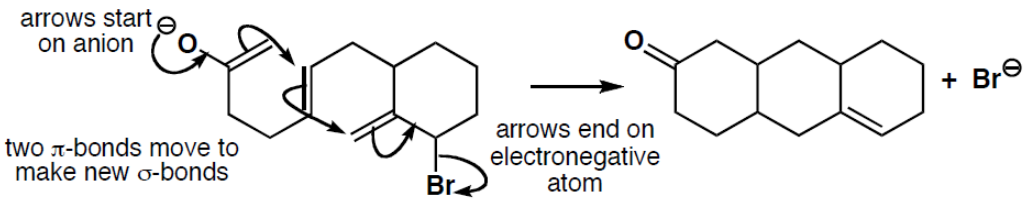




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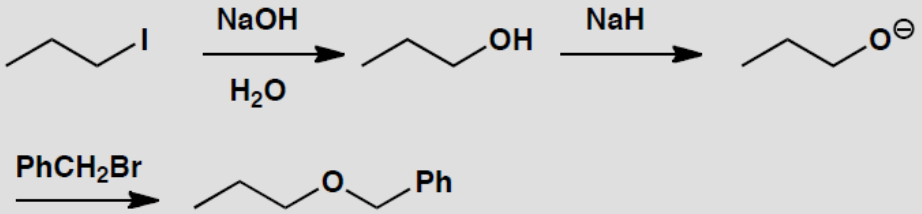
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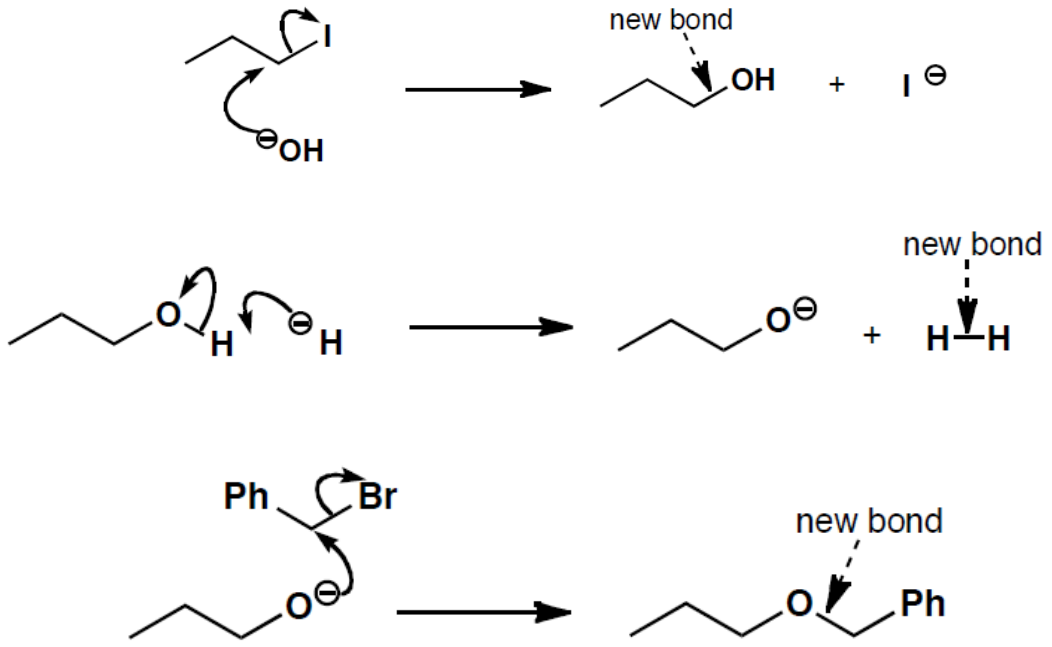
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Problem 5

Draw mechanisms for the reactions in the following sequence.





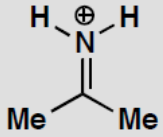
Problem 6

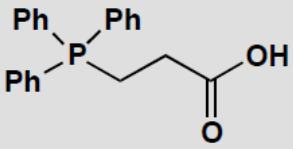
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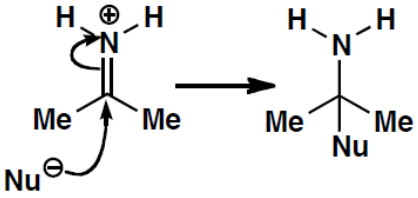
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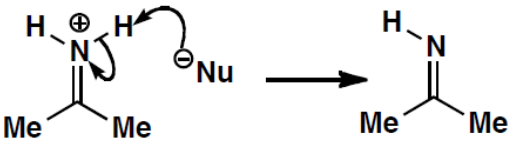
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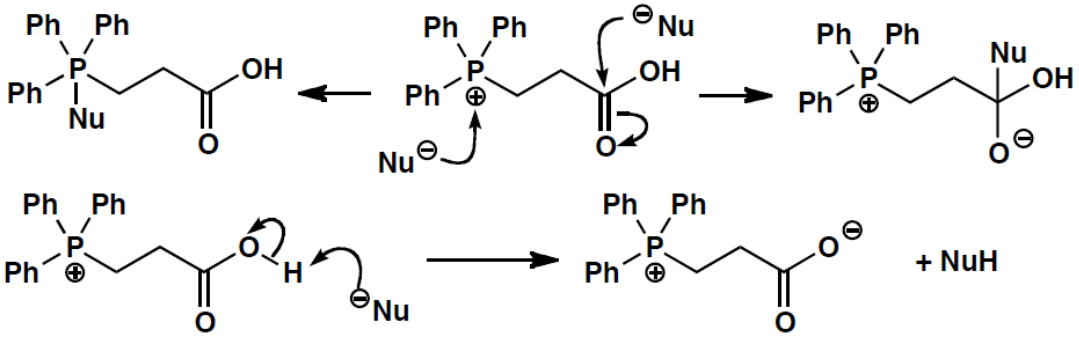
Each of these electrophiles could react with a nucleophile at one of (at least) two atoms. Identify these atoms and draw a mechanism and products for each reaction.











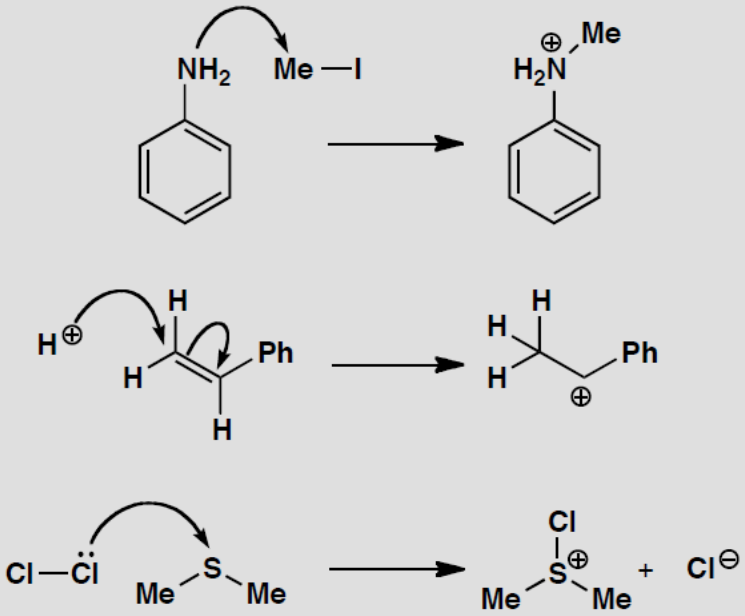
Problem 7

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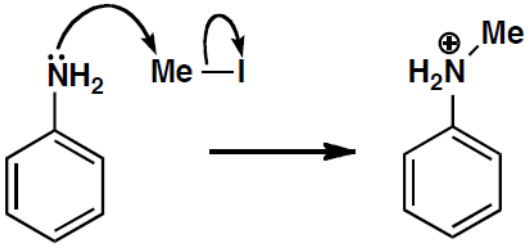
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These three reactions all give the products shown, but not by the mechanisms drawn! For each mechanism, explain what is wrong, and draw a better one.

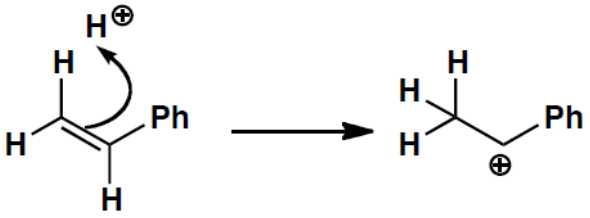


In the first reaction, the nucleophile is the amine and the electrophile is the methyl iodide, so the arrow is right in the sense that it starts on the nucleophile, where the electrons come from. However, we have stressed that a curly arrow should start on a representation of an electron pair, in other words on a lone pair, a bond or a negative charge. Here the arrow just starts on an atom: this is no good, and we must draw in the lone pair. The other problem is at the end of the arrow. It shows a new bond being formed to C, so unless a bond breaks then the C atom will have an impossible five bonds. We need another arrow to show the C–I bond breaking.

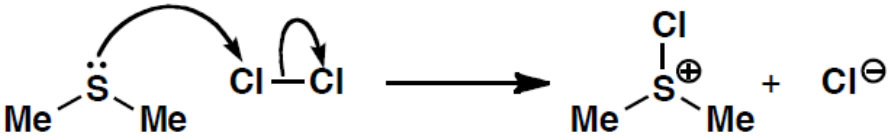


In the second reaction we form a cation by attack of a proton on an alkene. Which is the nucleophile? It can't be H^+: by definition a proton can't have a pair of electrons! The arrow must therefore start on the alkene and show the electrons moving towards the proton, not the other way round. The electrons come from the π bond, so the double bond is where we

start the arrow. We only need one arrow, because as the new C–H bond forms, the C atom at other end of the old π bond is left with only 6 electrons, and becomes positively charged.



The last reaction forms a new S–Cl bond. The arrow we have drawn starts on a lone pair, but if electrons are moving from Cl to S, surely the S will become negatively charged? The mistake is that the arrow is the wrong way round: the sulfur atom is the nucleophile, and the more electronegative chlorine is the electrophile. Start the arrow on the sulfur lone pair, and break the old Cl–Cl bond as the electrons arrive at Cl.



Problem 8

In your corrected mechanisms for problem 7, explain in each case which orbital is the HOMO of the nucleophile and which orbital is the LUMO of the electrophile.

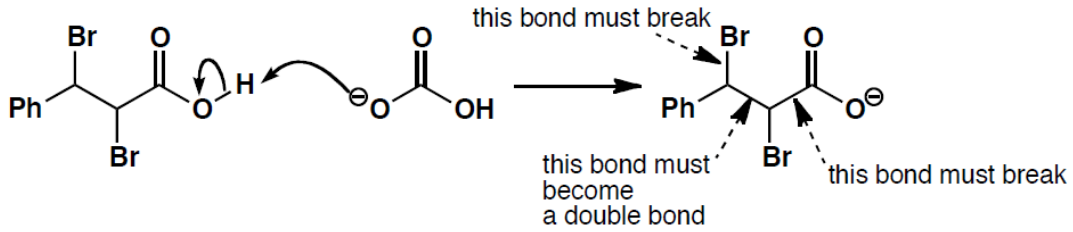
The nucleophilic amine of the first reaction uses its lone pair to form the new bond: the HOMO is the sp3 orbital containing this non-bonding electron pair. The empty orbital used by the electrophile must be an antibonding orbital, since as the electrons arrive there they cause the C–I bond to break: the LUMO is the C–I σ* orbital. When the alkene is protonated, it loses its π bond, and we have already pointed out that the electrons must come from this bond (that's why we start the arrow there). So the HOMO is the C=C π orbital. The LUMO is the only orbital the H+ ion has available: its empty 1s orbital. In the third reaction, the LUMO of the

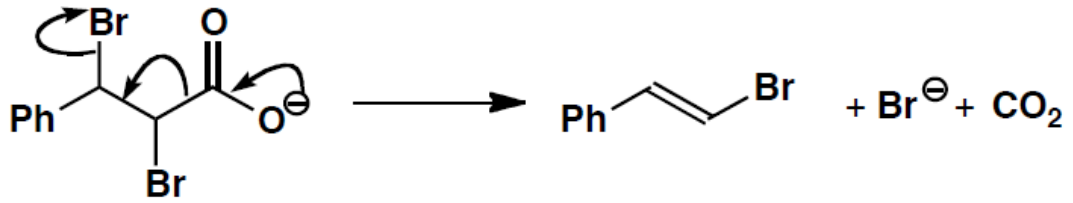
electrophile is easy to spot: it must be the Cl–Cl σ* orbital, since that is the bond that breaks. The nucleophile uses the lone pair on sulfur to react, so the HOMO is the non-bonding orbital occupied by this lone pair.

Problem 9

Draw a mechanism for the following reaction. (This is harder, but if you draw out the structures of the reactants first, and consider that one is an acid and one is a base, you will make a good start.)

$$PhCHBrCHBrCO_2H + NaHCO_3 \longrightarrow PhCH{=}CHBr + NaHCO_3$$

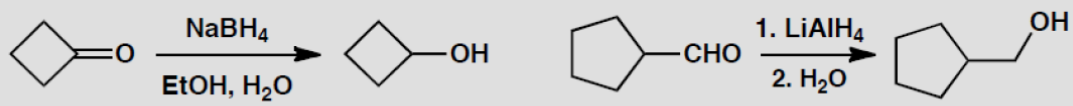




Tutorial 5

Problem 1

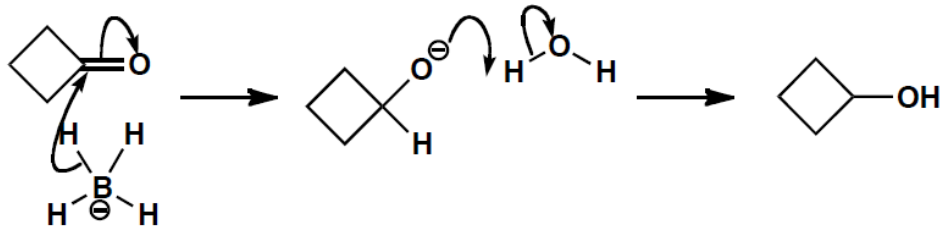
Draw mechanisms for these reactions:

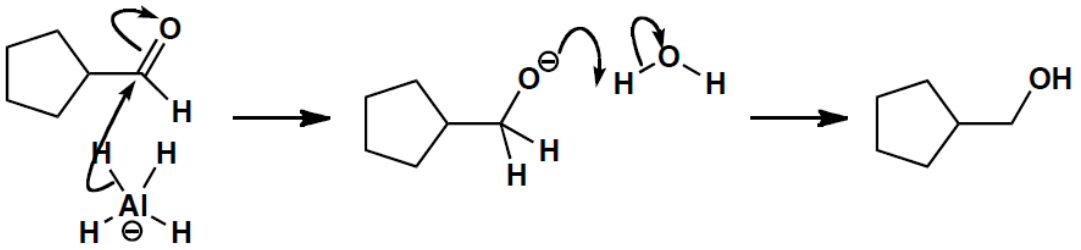


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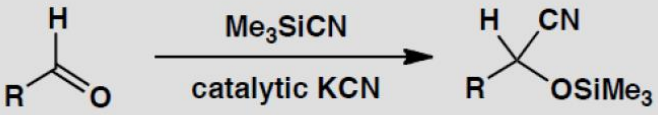
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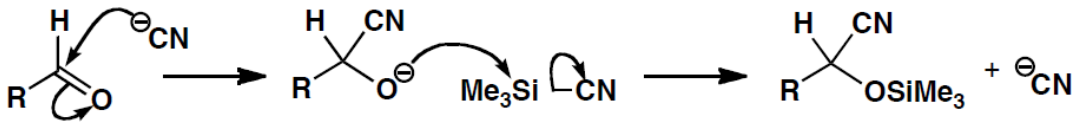


Problem 2

One way to make cyanohydrins is illustrated here. Suggest a detailed mechanism for the process.



The silyl cyanide is an electrophile while the cyanide ion in the catalyst is the nucleophile. Cyanide adds to the carbonyl group and the oxyanion product is captured by silicon, liberating another cyanide ion for the next cycle.



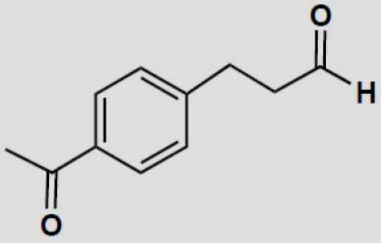
Problem 3

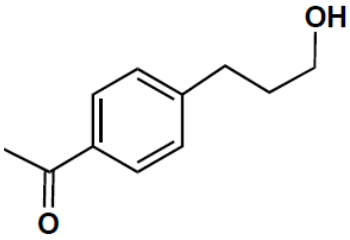
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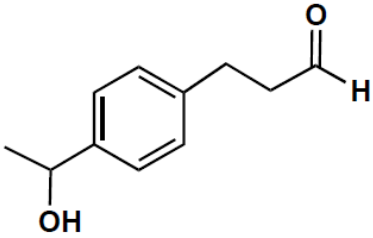
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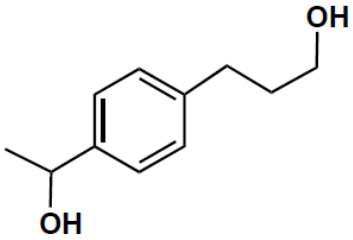
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There are three possible products from the reduction of this compound with sodium borohydride. What are their structures? How would you distinguish them spectroscopically, assuming you can isolate pure compounds?



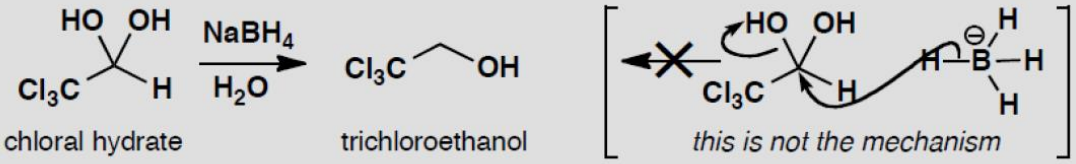






Problem 4

Trichloroethanol my be prepared by the direct reduction of chloral hydrate in water with sodium borohydride. Suggest a mechanism for this reaction. Take note that sodium borohydride does not displace hydroxide from carbon atoms!

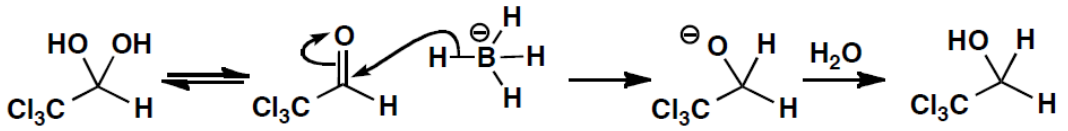


If sodium borohydride doesn't displace hydroxide from carbon atoms, then what does it do? We know it attacks carbonyl groups to give alcohols and to get trichloroethanol we should have to reduce chloral. Hemiacetals are in equilibrium with their carbonyl equivalents, so…

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Problem 5

What would be the products of these reactions? In each case give a mechanism to justify your prediction.



